

Quick Check

Lesson 7: Angles of Elevation and Depression: Solving Real Height-and-Distance Problems Across Kenya

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 7

A student stands 40 m from the base of a mast and measures the angle of elevation to the top as 50° . Let h be the height of the mast in metres. The 40 m side is next to the 50° angle, and h is opposite it. Which equation should be used to find h ?

A $\sin(50^\circ) = h/40$

B $\tan(50^\circ) = 40/h$

C $\cos(50^\circ) = h/40$

D $\tan(50^\circ) = h/40$

Which statement correctly describes an angle of depression?

- A** The angle measured downward from the horizontal to the line of sight
- B** The angle between the ground and the observer's feet
- C** The angle measured between the line of sight and the vertical line through the observer
- D** The angle measured upward from the horizontal to the line of sight

A person stands at the top of the Fort Jesus cliff in Mombasa, 20 m above sea level. The angle of depression to a dhow at sea level is 35° . What is the horizontal distance from the base of the cliff to the dhow, to 1 decimal place?

A 34.9 m

B 28.6 m

C 24.4 m

D 14.0 m

A visitor at Uhuru Park, Nairobi stands 15 m from the base of a flagpole. The angle of elevation to the top of the flagpole is 62° . What is the height of the flagpole, to 1 decimal place?

A 28.2 m

B 13.2 m

C 8.0 m

D 7.0 m

A surveyor in Kisumu stands 42 m (horizontal distance) from a five-storey building. The angle of elevation to the top of the building is 38° . What is the height of the building, to 1 decimal place?

A 53.8 m

B 32.8 m

C 25.9 m

D 33.1 m

A tourist stands at the top of a cliff 27 m above sea level. She sees a dhow at sea level at an angle of depression of 41° . Let d be the horizontal distance from the base of the cliff to the dhow. Which working correctly finds d ?

A $d = 27 \div \sin(41^\circ) \approx 41.2 \text{ m}$

B $d = 27 \times \tan(41^\circ) \approx 23.5 \text{ m}$

C $d = 27 \div \tan(41^\circ) \approx 31.1 \text{ m}$

D $d = 27 \times \cos(41^\circ) \approx 20.4 \text{ m}$

From the top of a cliff, the angle of depression to a boat on the sea is 41° . What is the angle of elevation from the boat to the person at the top of the cliff?

A 139°

B 90°

C 49°

D 41°

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).